

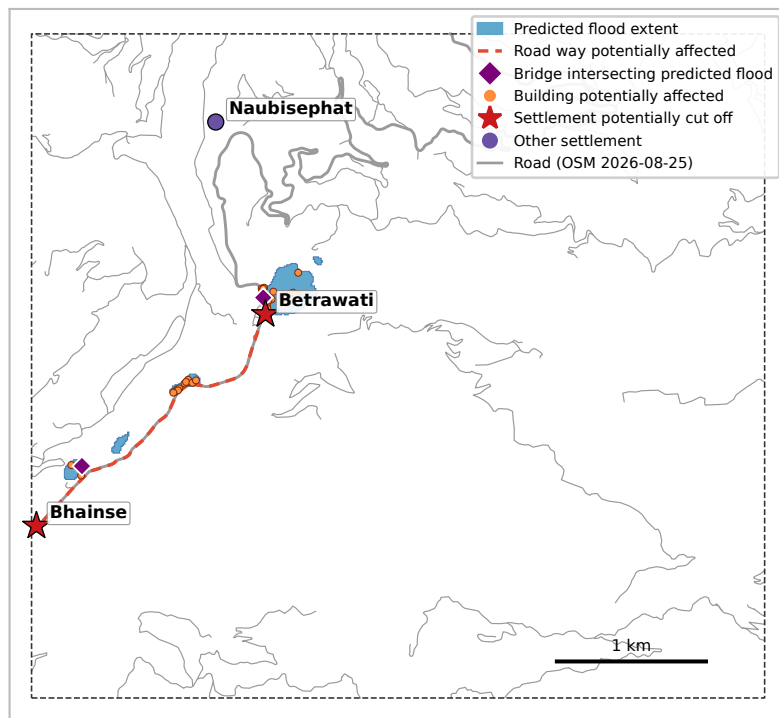
Trishuli Valley Flood Event · 26 August 2026

Post-event Sentinel-1: 28 August 2026 · Pre-event Sentinel-1: 16 August 2026, 04 August 2026 · relative orbit 85 ascending

AOI: Trishuli Valley (sub-AOI near Betrawati), Nepal (85.17–85.22° E, 27.95–27.99° N)

Indicators derived from an unvalidated model prediction and pre-event OpenStreetMap. Not confirmed damage or isolation.

1 FLOOD OVERVIEW



0.120 km²

Predicted flood extent

12.0 ha · 0.55 % of the analysed AOI pixels predicted as flood

Model prediction (class “flood”); not a confirmed flood area.

2 POTENTIAL INFRASTRUCTURE IMPACT

3

Road ways potentially affected

of 118 mapped road ways

189 m

Road length potentially affected

inside predicted flood; of 73.8 km mapped

2

Bridges intersecting predicted flood

of 10 mapped bridges

36

Buildings potentially affected

of 4,877 mapped buildings

3 CONNECTIVITY (SIMPLIFIED ROAD GRAPH)

3 settlements analysed; 2 potentially cut off: Betrawati, Bhainse.

3 road-graph edges blocked by predicted flood; main road network 31.1 km → 28.4 km (vehicle roads in AOI).

Betrawati (hamlet) Potentially cut off (simplified graph indicator)

reachable road network 31.13 km → 0.70 km; nearest road node 35 m

Bhainse (hamlet) Potentially cut off (simplified graph indicator)

reachable road network 31.13 km → 0.39 km; nearest road node 28 m

Naubisephat (hamlet) Already separate from main road network (before flood)

reachable road network 1.45 km → 1.45 km; nearest road node 187 m

Potentially cut off = disconnected in the simplified road graph after removing flood-affected edges.

This is a graph-based indicator, not confirmed real-world isolation.

4 METHODOLOGY

Sentinel-1 GRD → Kuro Siwo-trained U-Net → predicted flood extent → historical OSM infrastructure → spatial impact analysis → simplified road-network connectivity

Historical OSM: 2026-08-25T00:00:00Z (Overpass historical/attic snapshot, pre-event).

Model: Kuro Siwo-trained U-Net; 6-channel Sentinel-1 temporal input (VV/VH: post, pre1, pre2).

Prototype impact rules: roads ≥ 10 m inside predicted flood; bridges intersect predicted flood; buildings > 0.01 m² overlap. Not calibrated damage thresholds.

5 LIMITATIONS

- Trishuli flood prediction is unvalidated against ground truth in this project.
- Terrain-related SAR false positives are possible.
- Bridge intersection does not establish bridge damage or impassability.
- Potentially cut off is a simplified graph indicator, not confirmed isolation.
- Footpaths are excluded from the connectivity graph.
- The road graph is limited to the analysis AOI.

Data attribution

Sentinel-1 / Sentinel-2 (Copernicus free & open data): Contains modified Copernicus Sentinel data 2026, processed by the VantageQ team

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Kuro Siwo dataset (CC BY 4.0): Cite Bountos et al., NeurIPS 2024

OpenStreetMap / ohsome (ODbL 1.0): © OpenStreetMap contributors

Generated from VantageQ pipeline outputs; checkpoint sha256 0e28f6a44b774c5b...; OSM snapshot 2026-08-25T00:00:00Z.